

Algebra II with Trigonometry

Chapter 5.3

Olympiad Solutions (Gemini)

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[4401269_alg-2trig-h-chp-5-3-note-docx.pdf](#)

Solutions

1. Two triangles have the same area. The first triangle has side lengths of 5 and 8 with an included angle of 15° . The second triangle has side lengths of 4 and 10 with an included obtuse angle θ . What is the degree measure of θ ?

The area K of a triangle with side lengths a and b and included angle C is given by $K = \frac{1}{2}ab \sin C$. For the first triangle, the area is

$$K_1 = \frac{1}{2}(5)(8) \sin 15^\circ = 20 \sin 15^\circ. \text{ For the second triangle, the area is } K_2 = \frac{1}{2}(4)(10) \sin \theta = 20 \sin \theta.$$

Since the triangles have the exact same area, we equate the two expressions: $20 \sin \theta = 20 \sin 15^\circ \implies \sin \theta = \sin 15^\circ$.

Because θ is an obtuse angle ($90^\circ < \theta < 180^\circ$) and the sine function satisfies $\sin(180^\circ - x) = \sin x$, it follows that: $\theta = 180^\circ - 15^\circ = 165^\circ$.

2. Given that $\sin \theta + \cos \theta = \frac{1}{5}$ and that θ lies in Quadrant II, determine the exact value of $\sin \theta - \cos \theta$.

Squaring the given equation yields: $(\sin \theta + \cos \theta)^2 = \frac{1}{25}$

$$\sin^2 \theta + 2 \sin \theta \cos \theta + \cos^2 \theta = \frac{1}{25}$$

Using the Pythagorean identity $\sin^2 \theta + \cos^2 \theta = 1$, we find:

$$1 + 2 \sin \theta \cos \theta = \frac{1}{25} \implies 2 \sin \theta \cos \theta = -\frac{24}{25}.$$

We wish to evaluate $x = \sin \theta - \cos \theta$. Squaring this expression gives:
$$x^2 = (\sin \theta - \cos \theta)^2 = \sin^2 \theta - 2 \sin \theta \cos \theta + \cos^2 \theta = 1 - 2 \sin \theta \cos \theta.$$

Substituting the value of $2 \sin \theta \cos \theta$ obtained above: $x^2 = 1 - \left(-\frac{24}{25}\right) = \frac{49}{25}$.
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Because θ lies in Quadrant II, $\sin \theta > 0$ and $\cos \theta < 0$. Therefore, $\sin \theta - \cos \theta$ must be strictly positive. Taking the positive square root, we obtain: $\sin \theta - \cos \theta = \frac{7}{5}$.

3. Find the exact sum of the first 100 terms of the following sequence:
 $\cos(120^\circ) + \cos(240^\circ) + \cos(360^\circ) + \cos(480^\circ) + \dots$

The sequence of angles forms an arithmetic progression with a common difference of 120° . Let us evaluate the first few terms of the sequence:
 $\cos(120^\circ) = -\frac{1}{2} \quad \cos(240^\circ) = -\frac{1}{2} \quad \cos(360^\circ) = 1$

Because the cosine function is periodic with a period of 360° , the sequence repeats these three values indefinitely. The sum of any block of three consecutive terms is: $-\frac{1}{2} - \frac{1}{2} + 1 = 0$.

The sum of the first 100 terms can be split into 33 full blocks of 3 terms, plus one remaining term: $100 = 33 \times 3 + 1$.

The sum of the 33 full blocks is exactly 0. The single remaining term corresponds to the first term of the 34th block, which is $\cos(120^\circ) = -\frac{1}{2}$. Thus, the sum of the first 100 terms is $-\frac{1}{2}$.

4. The six vertices of a regular hexagon centered at the origin correspond to standard position angles $\theta_1, \theta_2, \dots, \theta_6$. If one of the vertices lies on the positive x -axis, mentally evaluate the product:
 $(1 - \sin \theta_1)(1 - \sin \theta_2)(1 - \sin \theta_3)(1 - \sin \theta_4)(1 - \sin \theta_5)(1 - \sin \theta_6)$

Since one vertex lies on the positive x -axis, the standard position angles are $0^\circ, 60^\circ, 120^\circ, 180^\circ, 240^\circ$, and 300° . Evaluating the sine of each angle gives

the respective set of values: $\{0, \frac{\sqrt{3}}{2}, \frac{\sqrt{3}}{2}, 0, -\frac{\sqrt{3}}{2}, -\frac{\sqrt{3}}{2}\}$.

The desired product is:

$$P = (1 - 0) \left(1 - \frac{\sqrt{3}}{2}\right) \left(1 - \frac{\sqrt{3}}{2}\right) (1 - 0) \left(1 + \frac{\sqrt{3}}{2}\right) \left(1 + \frac{\sqrt{3}}{2}\right).$$

By reordering the factors to pair the binomial conjugates, we gain a difference of squares: $P = 1 \cdot 1 \cdot \left[\left(1 - \frac{\sqrt{3}}{2}\right) \left(1 + \frac{\sqrt{3}}{2}\right)\right]^2$.

Simplifying the inner product: $1 - \left(\frac{\sqrt{3}}{2}\right)^2 = 1 - \frac{3}{4} = \frac{1}{4}$.

Therefore, the total product is: $P = \left(\frac{1}{4}\right)^2 = \frac{1}{16}$.

5. A circular sector has an acute central angle θ . The ratio of the area of the triangle formed by its two bounding radii and the chord connecting their endpoints, to the area of the entire sector, is exactly $\frac{3}{\pi}$. Find the degree measure of θ .

Let r be the radius of the circle, and let the central angle θ be measured in radians. The area of the triangle formed by the two radii and the chord is $A_{\text{triangle}} = \frac{1}{2}r^2 \sin \theta$. The area of the circular sector is $A_{\text{sector}} = \frac{1}{2}r^2\theta$.

The given ratio is: $\frac{A_{\text{triangle}}}{A_{\text{sector}}} = \frac{\frac{1}{2}r^2 \sin \theta}{\frac{1}{2}r^2\theta} = \frac{\sin \theta}{\theta}$.

We are given that this ratio is $\frac{3}{\pi}$, yielding the equation: $\frac{\sin \theta}{\theta} = \frac{3}{\pi}$.

Notice that by inspection, substituting $\theta = \frac{\pi}{6}$ gives: $\frac{\sin(\pi/6)}{\pi/6} = \frac{1/2}{\pi/6} = \frac{3}{\pi}$.

Because the function $f(\theta) = \frac{\sin \theta}{\theta}$ is strictly decreasing on the interval $(0, \pi/2)$, $\theta = \frac{\pi}{6}$ is the unique acute angle that satisfies this equation. Converting $\frac{\pi}{6}$ radians to degrees, we find that the degree measure is 30° .

6. If θ is an acute angle in standard position such that $\sec \theta + \tan \theta = 4$, compute the exact value of $\sin \theta$.

Recall the fundamental Pythagorean identity: $\sec^2 \theta - \tan^2 \theta = 1$.

This can be factored as a difference of squares:

$$(\sec \theta - \tan \theta)(\sec \theta + \tan \theta) = 1.$$

Substituting the given equation $\sec \theta + \tan \theta = 4$ into the identity yields:

$$4(\sec \theta - \tan \theta) = 1 \implies \sec \theta - \tan \theta = \frac{1}{4}.$$

We now have a straightforward linear system in terms of $\sec \theta$ and $\tan \theta$: 1)

$$\sec \theta + \tan \theta = 4 \quad 2) \quad \sec \theta - \tan \theta = \frac{1}{4}$$

Adding the two equations eliminates $\tan \theta$:

$$2 \sec \theta = 4 + \frac{1}{4} = \frac{17}{4} \implies \sec \theta = \frac{17}{8}. \text{ Since } \sec \theta = \frac{1}{\cos \theta}, \text{ we have } \cos \theta = \frac{8}{17}.$$

Subtracting the second equation from the first eliminates $\sec \theta$:

$$2 \tan \theta = 4 - \frac{1}{4} = \frac{15}{4} \implies \tan \theta = \frac{15}{8}.$$

Using the definition $\tan \theta = \frac{\sin \theta}{\cos \theta}$, we solve for $\sin \theta$:

$$\sin \theta = \tan \theta \cos \theta = \left(\frac{15}{8}\right) \left(\frac{8}{17}\right) = \frac{15}{17}.$$